

# LECTURE-21

⇒ ISSUES IN ML: Underfit & Overfit.

|       |      | Training Error |                                                      |
|-------|------|----------------|------------------------------------------------------|
|       |      | Low            | High                                                 |
| Error | Low  | Ideal          | issue in code → OR complex training, simple testing. |
|       | High | Overfit        | Underfit                                             |

- ★ High biasness towards class causes Underfitting
- ★ High variance causes Overfitting.

⇒ When we've multiple classes, we'll use "One-Hot-Encoding"  
Classes: Orange, Mango, Apple.

| $x_1$ | $x_2$ | Label  | $y$ | One hot encoded |
|-------|-------|--------|-----|-----------------|
| 2     | 1     | Apple  | 1   | 001             |
| 3     | 2     | mango  | 2   | 010             |
| 6     | 5     | orange | 3   | 100             |
| 5     | 7     | orange | 3   | 100             |
| 9     | 3     | apple  | 1   | 001             |

\* The output ' $y$ ' in  
qls will also be  
in three bit form  
like  $y = 010$ .